After running batch Thompson sampling, we can plot the progress made by the policy against other policies we have learned, as shown in Figure A.16. Here, Thompson sampling is able to make significant progress only after the first batch of queries.

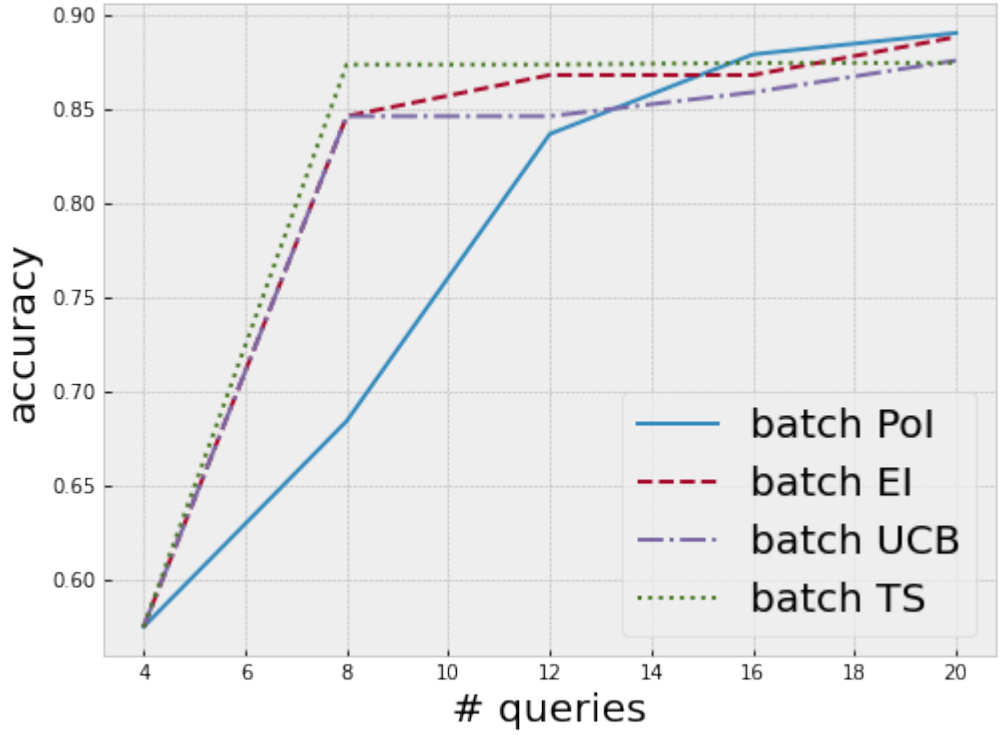


Figure A.16 Progress made by batch Thompson sampling in the hyperparameter tuning example. The policy makes significant progress only after the first batch of queries.

A.7.2 Optimizing airplane designs

This exercise provides an objective function that simulates this process of benchmarking the performance of an airplane design. The code is provided in `CH07/04 - Exercise 2.ipynb`. There are multiple steps:

1. Implement the objective function that simulates the performance benchmarking.
   The code for the objective function is already provided, so we simply copy and paste it in our program:

```
def flight_utility(X):
    X_copy = X.detach().clone()
    X_copy[:, [2, 3]] = 1 - X_copy[:, [2, 3]]
    X_copy = X_copy * 10 - 5

    return -0.005 * (X_copy**4 - 16 * X_copy**2 + 5 * X_copy).sum(dim=-1) + 3
```

2. Implement a Gaussian process (GP) model with a constant mean function and a Matérn 2.5 kernel with an output scale implemented as a `gpytorch.kernels.ScaleKernel` object.
   The class implementation of this GP model is mostly the same as before, except for that we need to specify the correct number of dimensions in the ARD kernel:

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